

Bank Demo

1. Objective

To develop a Java program to demonstrate **method overriding and dynamic polymorphism** using the RBI, SBI, ICICI, and PNB classes, and to display the rate of interest based on the bank selected by the user.

2. Algorithm

1. Start the program.
2. Create a Customer class with customer name and customer ID.
3. Create an Account class with account number and balance.
4. Create the RBI base class with the `getRateOfInterest()` method.
5. Create SBI, ICICI, and PNB classes that inherit from RBI.
6. Override the `getRateOfInterest()` method in each child class with its respective interest rate.
7. Read the bank name from the user using Scanner.
8. Declare an RBI reference variable.
9. Assign the appropriate bank object to the RBI reference based on the user's input.
10. Call the overridden `getRateOfInterest()` method.
11. Display the rate of interest.
12. If an invalid bank name is entered, display "Invalid Bank Name".
13. Stop the program.

3. UML Class Diagram

RBI + getRateOfInterest()
Extends

SBI + getRateOfInterest()	ICICI + getRateOfInterest()	PNB + getRateOfInterest()
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Customer - name : String - customerId : displayCustomer()	Account - accountNumber : int - balance : double + displayAccount()
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4. Source Code

```
import java.util.Scanner;

// Customer class
class Customer{
    String name;
    int customerId;

    // Constructor
    Customer(String name, int customerId){
        this.name = name;
        this.customerId = customerId;
    }

    // Display customer details
    void displayCustomer(){
        System.out.println("Customer Name : " + name);
        System.out.println("Customer ID  : " + customerId);
    }
}

// Account class
class Account{
    int accountNumber;
    double balance;

    // Constructor
    Account(int accountNumber, double balance){
        this.accountNumber = accountNumber;
```

```

        this.balance = balance;
    }

    // Display account details
    void displayAccount(){
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Balance      : " + balance);
    }
}

// RBI is the Base Class
class RBI{
    // Minimum interest rate fixed by RBI
    double getRateOfInterest(){
        return 4.0;
    }
}

// SBI inherits RBI
class SBI extends RBI{
    // SBI provides its own interest rate
    @Override
    double getRateOfInterest(){
        return 7.0;
    }
}

// ICICI inherits RBI
class ICICI extends RBI{
    // ICICI provides its own interest rate
    @Override

```

```

double getRateOfInterest(){
    return 6.5;
}
}

// PNB inherits RBI
class PNB extends RBI{
    // PNB provides its own interest rate
    @Override
    double getRateOfInterest(){
        return 6.0;
    }
}

// Main class
public class BankDemo{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        // Taking bank name from user
        System.out.print("Enter the Bank name to find the rate of Interest : ");
        String bankName = sc.nextLine();
        // RBI reference variable
        RBI rbi;
        // Dynamic polymorphism
        // The RBI reference can refer to different child objects
        if(bankName.equalsIgnoreCase("RBI")){
            rbi = new RBI();
        }
        else if(bankName.equalsIgnoreCase("SBI")){

```

```
        rbi = new SBI();
    }
    else if(bankName.equalsIgnoreCase("ICICI")){
        rbi = new ICICI();
    }
    else if(bankName.equalsIgnoreCase("PNB")){
        rbi = new PNB();
    }
    else{
        System.out.println("Invalid Bank Name");
        sc.close();
        return;
    }
    // Calling overridden method
    // Method is decided at runtime based on the object
    System.out.println("RBI rate of interest is : " + rbi.getRateOfInterest() + "%");
    sc.close();
}
}
```

5. Output

Sample Input 1:

Enter the Bank name to find the rate of Interest : SBI

Output:

RBI rate of interest is : 7.0%

Sample Input 2:

Enter the Bank name to find the rate of Interest : ICICI

Output:

RBI rate of interest is : 6.5%

Sample Input 3:

Enter the Bank name to find the rate of Interest : PNB

Output:

RBI rate of interest is : 6.0%

Sample Input 4:

Enter the Bank name to find the rate of Interest : ABC

Output:

Invalid Bank Name